

Lab 1: DATA PREPARATION

DUE: 7:50am 09/03/2025

Submission: mlearning

- **Lab1_YOURNAME.ipynb**
- Must include all data file (.csv,...) and .ipynb in your submission
- .csv and .ipynb must be in the same folder (pd.read_csv('data.csv'))
- No use of LOCAL libraries/folders
- NO LATE submission accepted

```
import pandas as pd

df = pd.read_csv('data.csv')

print(df.head())
```

Use 'dataset.csv'. Notes: All histograms must be labeled.

- Find the number of missing data in all columns. What's the missing percentage? Hint: use **df.isnull()** or **df.isna()**. Show the first 10 lines of the output.
- Use the original data. Fill in the missings with [column] **mean**. Hint: **fillna()**. Show the first 10 lines of the output.
- Use the original data. Fill in the missings with [column] **median**. Show the first 10 lines of the output.
- Use the original data. Fill in the missings with [column] **mode**. Show the first 10 lines of the output.
- What is the difference between mean, median and mode?
- Use the original data. Drop columns with a missing value rate higher than **30%**. Show the first 10 lines of the output.

- g) Use the original data. Drop rows with missing value rate higher than **40%**. Show the first 10 lines of the output.
- h) Use the original data. Normalize data **columns** using minmax normalization. Show the first 10 lines of the output.
- i) Outlier removal: Fill in the missings with **mode**. Use the First Column, Drop the outlier rows with $2 * \text{standard deviation}$,